



Technical Deck - April 2026

Dr. Viknish Krishnan-Kutty
Founder / CEO

Problem Statement

FBS is the gold standard for growing cells – used in research, regenerative medicine, cultivated meat, and biologics.

But it comes from blood drawn from cow fetuses during slaughter, making it unethical, highly variable, and supply-constrained. Worse, it costs more than \$1000 per liter and has performance unpredictability that leads to failed experiments or failed products.

The industry has tried alternatives for decades, but most are either underperforming or not scalable.

The market is desperate for a solution!

1) Technical Vision

Biology is controlled by signals—not just nutrients

- Conventional cell culture rely on Fetal Bovine Serum (FBS), a crude, undefined mixture
- FBS delivers both:
 - Nutritional support
 - Biological signalling cues (poorly understood, highly variable)
- ➔ While both are important, the true drivers of cell behaviour are extracellular signalling systems. This can be seen from the development specialised cells in a developing organism
- Cells communicate and regulate function via extracellular vesicles (EVs). These EVs carry:
 - Proteins, RNAs and Regulatory molecules

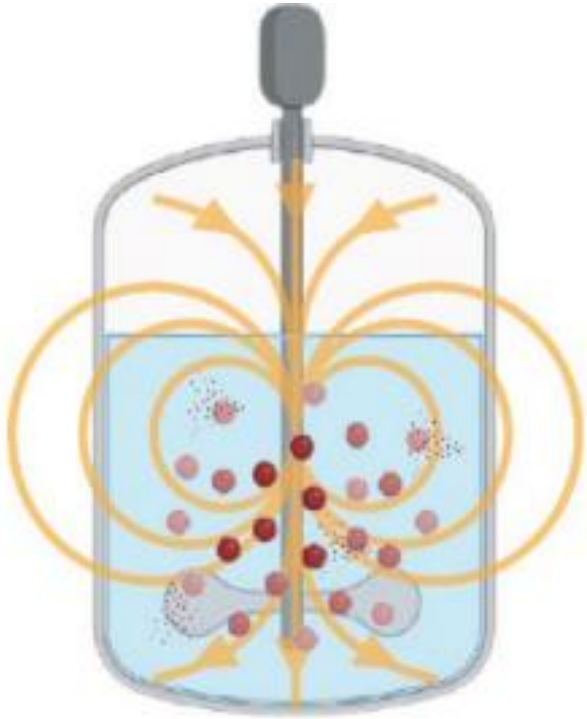
1) Technical Vision

Engineering cells to produce functional EV-based inputs

- Proprietary stimulation system induces cells to release bioactive EVs at scale
- Output: a cell-conditioned, signal-rich biological input layer
- Designed to replace undefined serum with:
 - Higher consistency
 - Tunable biological function
 - Scalable production
- EV based signalling is:
 - Programmable (in principle)
 - Largely untapped in industrial biology

1) Technical Vision – More about EVs

Extracellular Vesicles - 2013 Nobel Prize in Physiology or Medicine



- 30nm – 10um; 6000x smaller than hair
- Cell signaling and communication
- Stressed cells – Stress signals
- Happy cells – Happy signals
- Application emerging in therapy, skincare, drug delivery, etc

1) Technical Vision

Product strategy: Enter with FBS replacement, expand across industries

- **Phase 1:** Replace FBS in cell culture (research, biopharma, cultivated meat)
- **Phase 2:** Extend EV platform into:
 - Skincare actives (regenerative signalling; we already have data that supports collagen increase. See in later slides)
 - Therapeutics (targeted biological modulation)
 - Advanced growth systems

1) Technical Vision

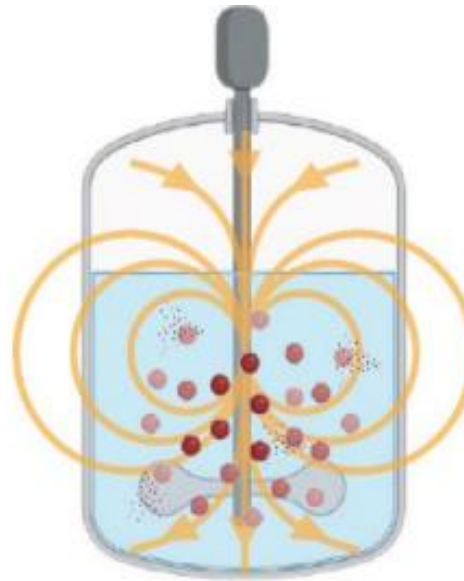
End state: A programmable extracellular signalling platform

- Move from:
 - Feeding cells → “instructing cells”
- Build a foundation for:
 - Predictable biology
 - Cross-industry applications
 - Data-driven optimisation of cellular systems

2) Platform Technology Overview

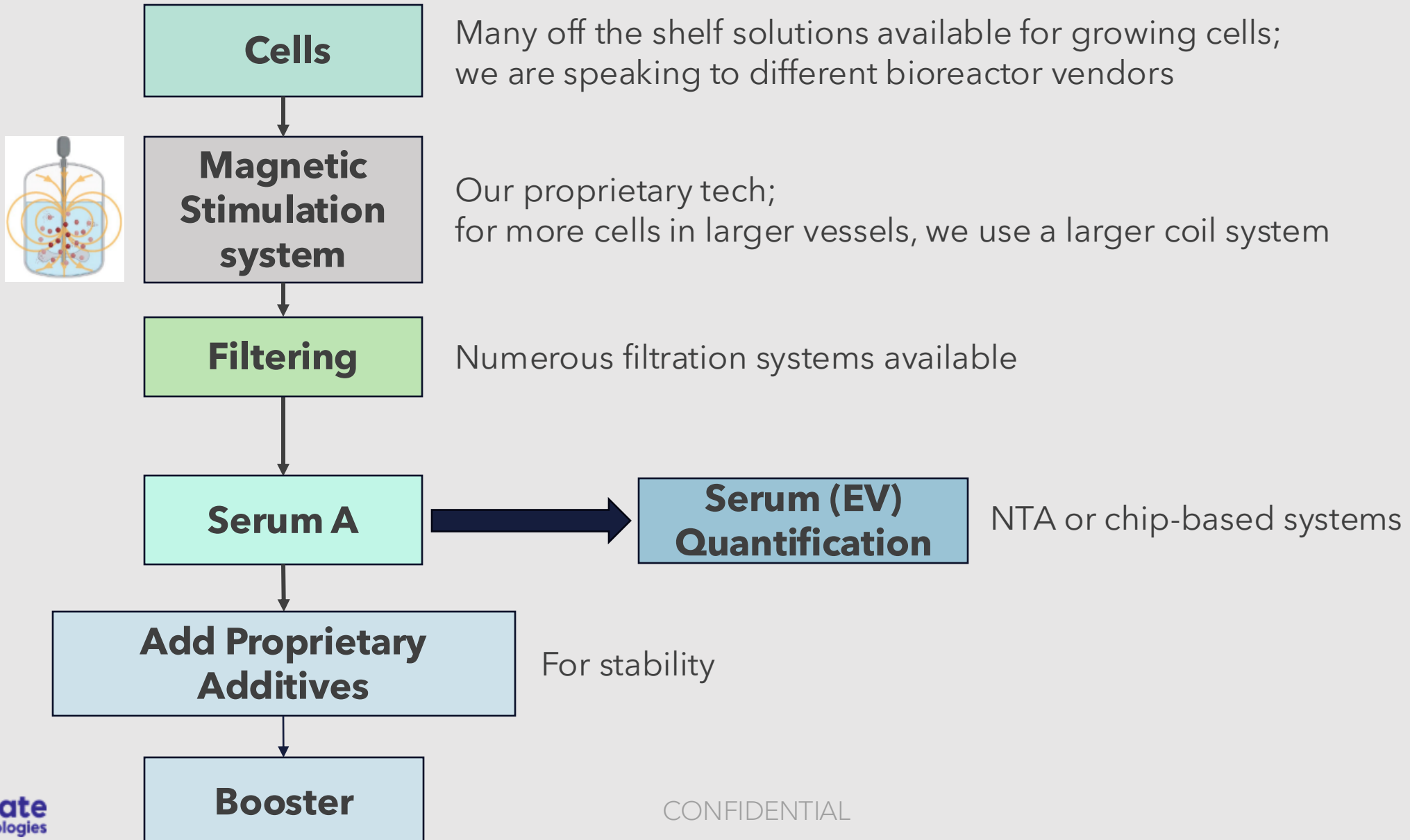


**Grow cells of interest
(Any preferred format)**

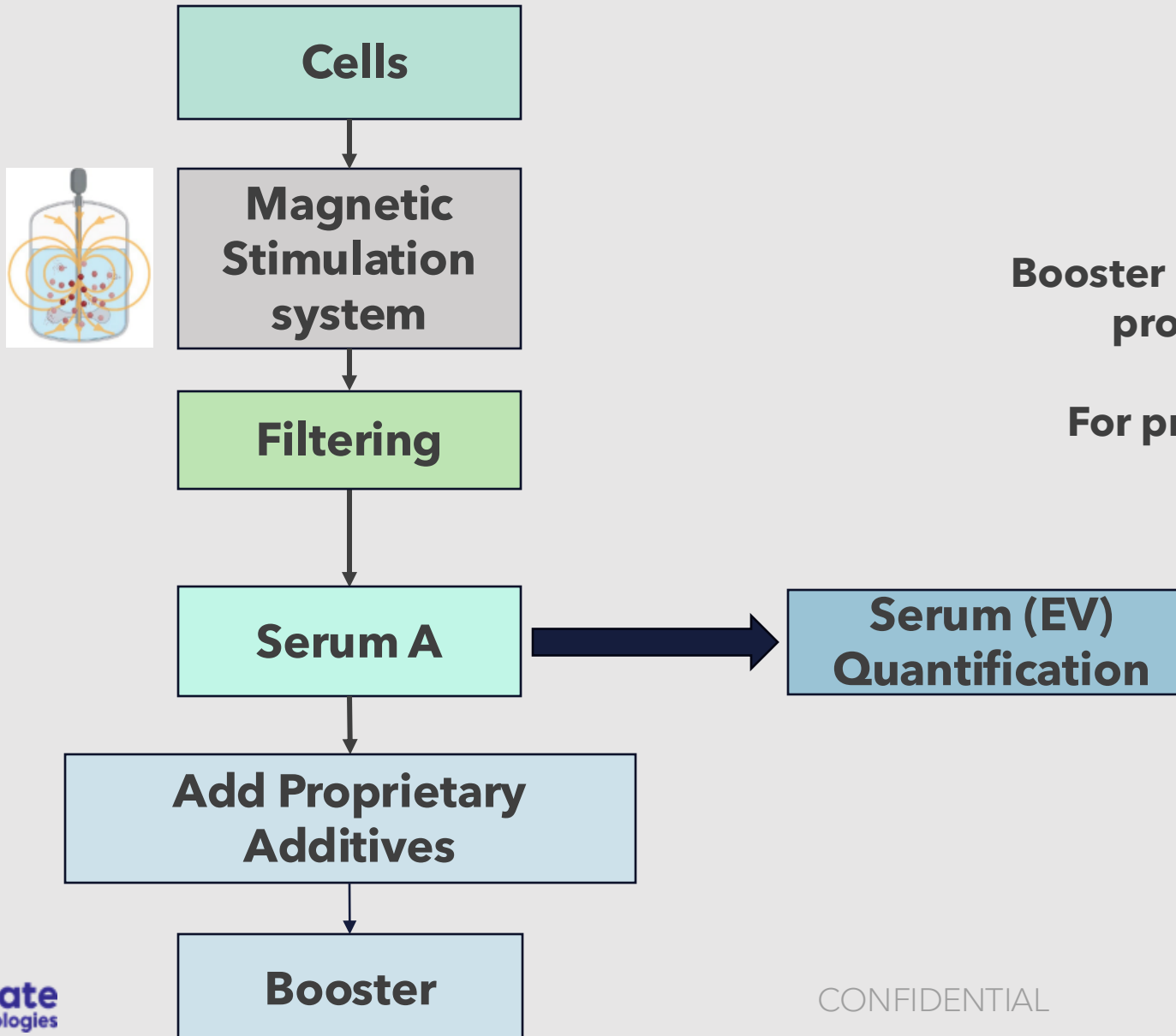


**Collect serum
(BOOSTER™)**

2) Platform Technology Overview



2) Platform Technology Overview



Important Note:

Booster Basic is a subset of Booster. As such, the process is contained in this schematic.

For proprietary reasons, the detailed steps involved in Booster Basic are not shared.

3) Mechanism of Action (MoA)

- Biological rationale:
 - How magnetic stimulation → cellular response
 - How this alters EV release / composition
- Supporting literature (even indirect)

3) MOA-Biological rationale of technology

- Magnetic stimulation induces mitohormesis (mitochondrial survival adaptations) promoting mitochondriogenesis, reducing basal apoptosis, increases telomere length, and secretome response output hence improving cell growth (1-3)
- Specifically, the magnetic stimulation activates a calcium-mitochondrial axis (Transient receptor potential canonical 1 channel (TRPC-1)), stimulating mitochondrial respiration and associated with PGC-1 α transcriptional co-activation which bring about all cell and tissue related processes (1-3)
- Magnetic stimulation through these specific mechanisms has been shown to promote myogenesis, neurogenesis, chondrogenesis, cancer, regeneration and other cell- based repair functions (4-9)
- PEMF induced magnetic stimulation increases physiological secretome output that contains many factors including Extracellular vesicles (EVs), proteins, lipids, metabolites and other bio-actives. These EVs have multiple potential from cell culture media , cell-based repair, therapy, regeneration, anti-cancer and cure for pathology and disease (10-12)

1-Wong et al., 2022; Kurth et al., 2020
2-Yap et al 2019
3-Scheele et al 2009

4-Parate et al., 2021
5-Celik et al ., 2019
6-TT Madan et al., 2021

7-Kurth et al., 2020
8-Alex et al., 2024
9-Alex et al., 2022

10-Kim et al 2022
11-Mohmadipoor et al 2023
12-Mc Laughlin et al., 2022

3) MOA-Biological rationale of technology-Key References

Core PEMF Technology References

1-Brief exposure to directionally-specific pulsed electromagnetic fields stimulates extracellular vesicle release and is antagonized by streptomycin: A potential regenerative medicine and food industry paradigm. BIOMATERIALS 287. ScholarBank@NUS Repository.

<https://doi.org/10.1016/j.biomaterials.2022.121658>

[2-Franco-Obregón et al. 2023.](#) The Developmental Implications of Muscle-Targeted Magnetic Mitohormesis: A Human Health and Longevity Perspective. Bioengineering 10 (8) : 956-. ScholarBank@NUS Repository. <https://doi.org/10.3390/bioengineering10080956>

3 Parate et al., 2020, Pulsed electromagnetic fields potentiate the paracrine function of mesenchymal stem cells for cartilage regeneration. STEM CELL RESEARCH & THERAPY 11 (1) : 46. ScholarBank@NUS Repository. <https://doi.org/10.1186/s13287-020-1566-5>

4-Alfredo et al., 2020, Cell-Derived Vesicles as TRPC1 Channel Delivery Systems for the Recovery of Cellular Respiratory and Proliferative Capacities, Adv. Biosys. 2020, 4, 2000146, DOI: 10.1002/adbi.202000146

PEMF- Mitohormesis, Regeneration and Inflammatory disorders

1-Alfredo et al., 2023, The Developmental Implications of Muscle-Targeted Magnetic Mitohormesis: A Human Health and Longevity Perspective, Bioengineering 2023, 10, 956. <https://doi.org/10.3390/bioengineering10080956>

2-Magnetic mitohormesis: A non-invasive therapy for inflammatory disorders? Biocell 2023

3-Celik et al., 2020 Directionalities of magnetic fields and topographic scaffolds synergise to enhance MSC chondrogenesis, [m5G;November 6, 2020;14:36] <https://doi.org/10.1016/j.actbio.2020.10.039>

4-Pulsed electromagnetic fields potentiate the paracrine function of mesenchymal stem cells for cartilage regeneration Parate et al. Stem Cell Research & Therapy (2020) 11:46 <https://doi.org/10.1186/s13287-020-1566-5>

5-Madan et al 2021, PULSED ELECTROMAGNETIC FIELDS SYNERGIZE WITH GRAPHENE TO ENHANCE DENTAL PULP STEM CELL-DERIVED NEUROGENESIS BY SELECTIVELY TARGETING TRPC1 CHANNELS European Cells and Materials Vol. 41 2021 (pages 216-232) DOI: 10.22203/eCM.v041a16

Extracellular Vesicles (EVs), Secretome, disease and Therapeutic applications

1- Song et al., 2019, Secretomics to Discover Regulators in Diseases., Int. J. Mol. Sci. 2019, 20, 3893; doi:10.3390/ijms20163893

2-Schank et al ., 2026, Extracellular Vesicles: A Comprehensive Review of Their Origins, Functions, and Therapeutic Potential Biomedicine 2026, 14, 495 <https://doi.org/10.3390/biomedicine14030495>

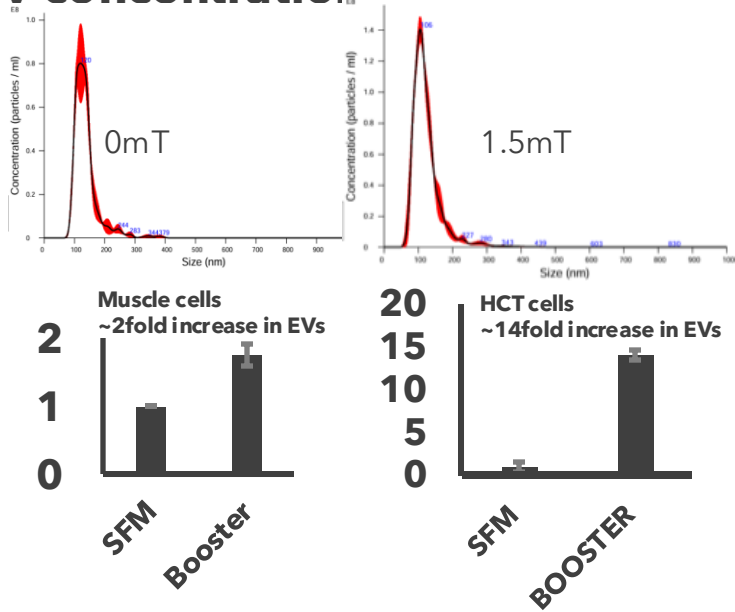
3.Kim et al., 2022, Secretome of Stem Cells: Roles of Extracellular Vesicles in Diseases, Stemness, Differentiation, and Reprogramming Tissue Eng Regen Med (2022) 19(1):19-33, <https://doi.org/10.1007/s13770-021-00406-4>

4) Product Characterisation

- Composition:
 - EV concentration, size distribution
 - Key proteins / growth factors (if known)
 -
- Analytical methods:
 - NTA, proteomics, RNA profiling, etc.
- Batch-to-batch consistency data

4) Product Characterisation

EV concentration size distribution through NTA



Proteomics summary

- Proteome analysis in general shows positive EV, secretome markers and specifically that supports cell growth, proliferation and reduced apoptosis
- Before magnetic stimulation, only Protein Tyrosine kinase signaling related proteins were identified suggesting its role in cell survival and baseline growth
- After magnetic stimulation a number of markers were upregulated involved in **cytokinesis, metabolic processes, ECM adhesion matrix assembly, G-protein coupled receptor signaling, cytoskeletal remodeling and Cell Growth/development and cysteine peptidases down regulated**
- The proteins that are upregulated suggest that they specifically have effect on recipient cell population and qualifies the composition of EVs secretome proteins from donor populations that are stimulated after stimulation

- Size distribution from 30-150nm EV populations observed
- Based on statistics the quantity of total EV population increases after PEMF stimulation. This is dependent on cell type used with PEMF stimulation (some instance 2 folds in muscle cells 14 folds in carcinoma cells after EV enrichment based on NTA analysis)
- Before magnetic stimulation with enriched EVs through NTA analysis exhibit dichotomous (two) peak distribution and after stimulation this changes in to single peak distribution

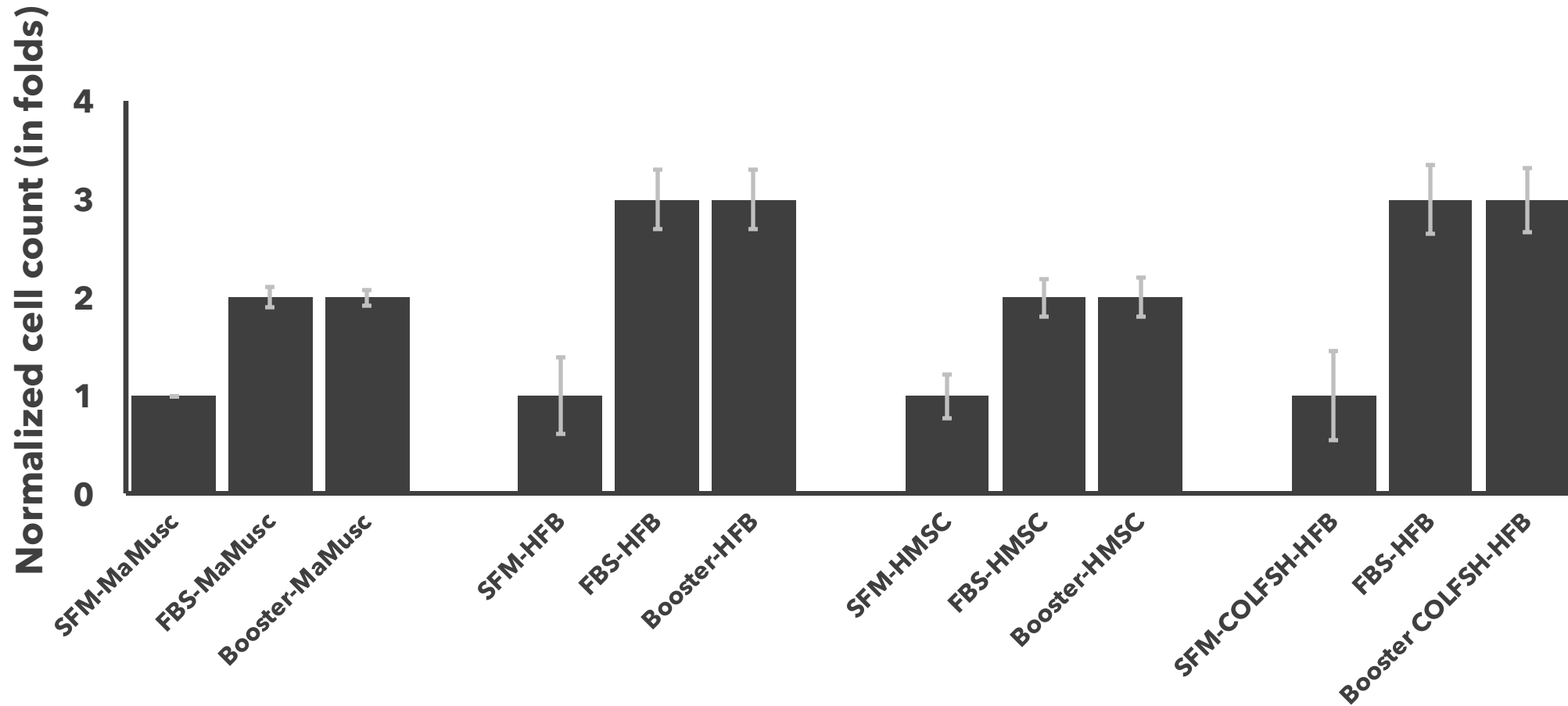
5) Performance Validation

- . Head-to-head vs FBS
 - Cell growth rates
 - Viability
 - Morphology
 - Functional assays (viral titre, differentiation, etc.)
- . Multiple cell types

5) Performance comparable across multiple cell types

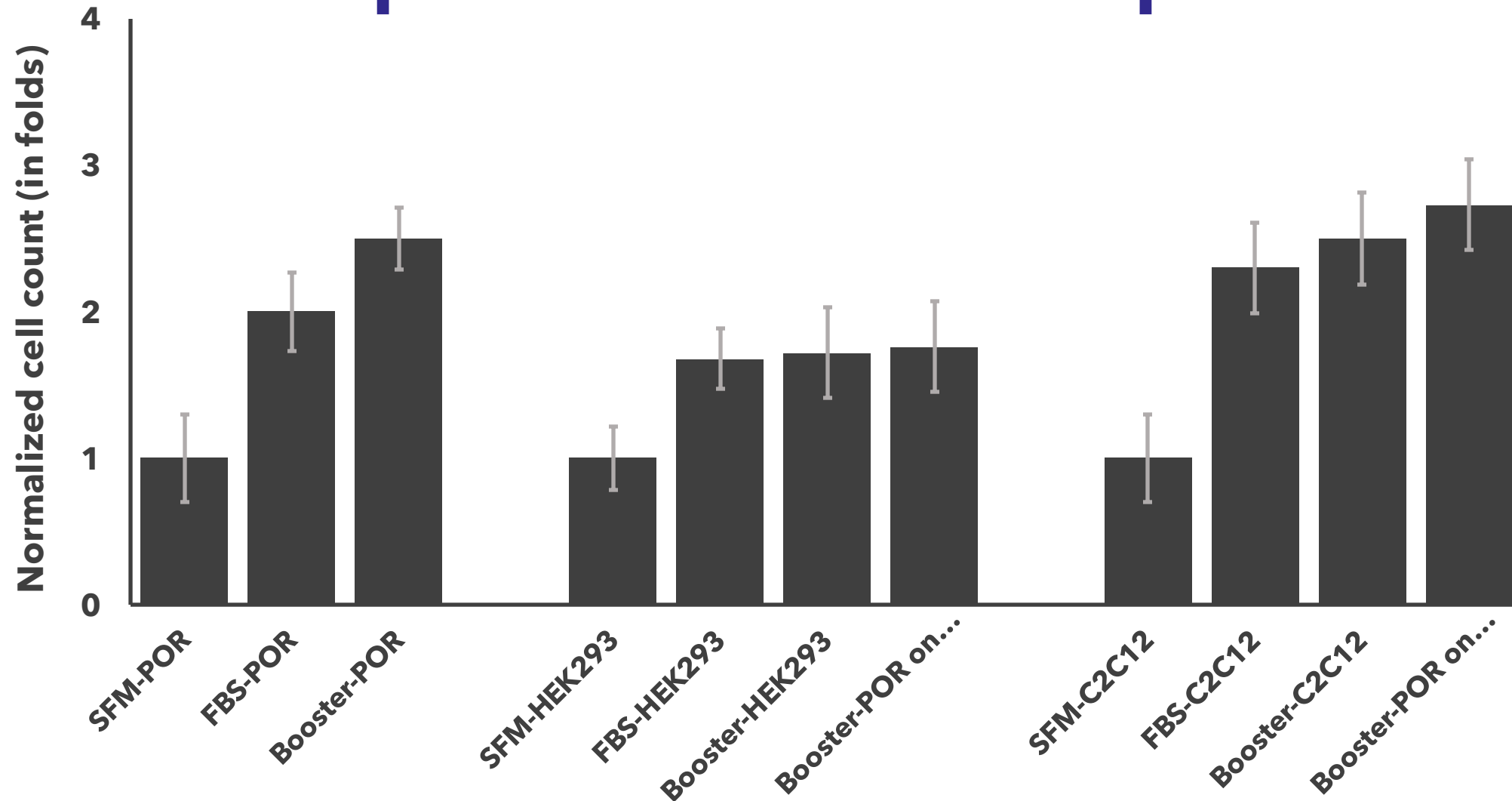
- Booster is generated through inducing secretome that contain EVs and multiple factors
- These factors generated from same donor or different donor cell type, has effect in recipient cells through autocrine or paracrine manner supporting Booster activity comparable to that of FBS
- Booster is comparable to FBS across multiple cell lines and primary cell types from different lineages
- Booster's performance is comparable with FBS across ten generations with consistent, reproducible functional capability in each instance
- Moreover, growth kinetics studies show Booster and FBS has similar consistency across four-day period with similar doubling time

5) Booster from same or different donor cell type stimulates recipients in autocrine or paracrine manner



MaMusc-Mammalian Muscle
HFB-Human Adult Fibroblast
HMSC-Human Mesenchymal stem cells
COLFSH-FISH cells

5) Booster from same or different donor cell type stimulates recipients in autocrine or paracrine manner

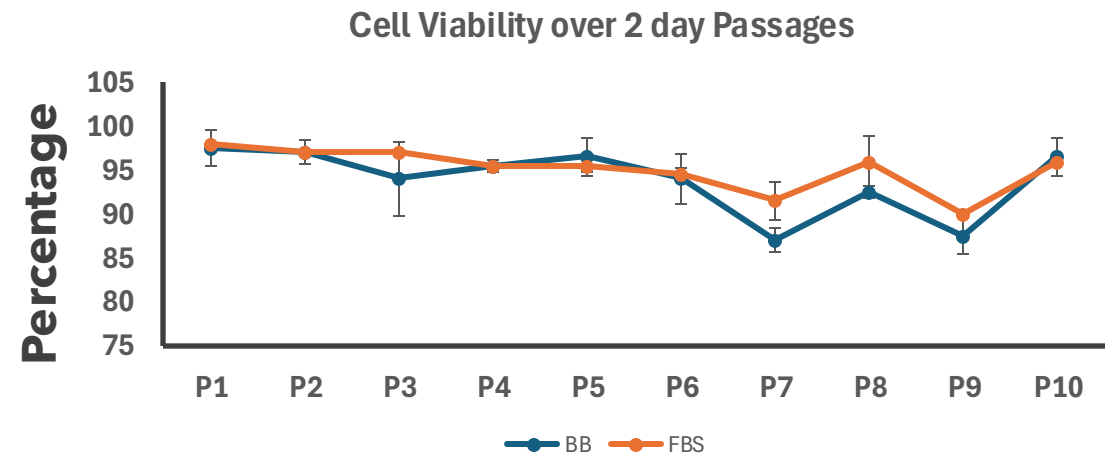
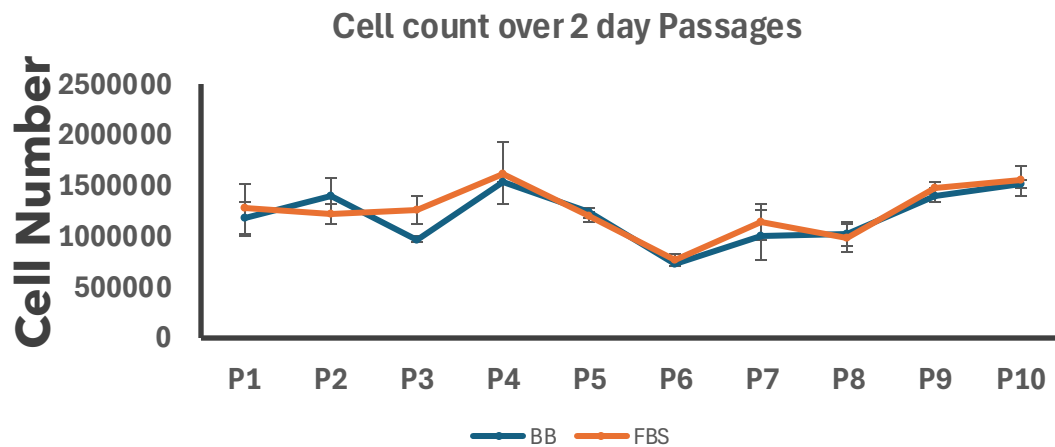


POR-Porcine
HEK-Human Embryonic kidney

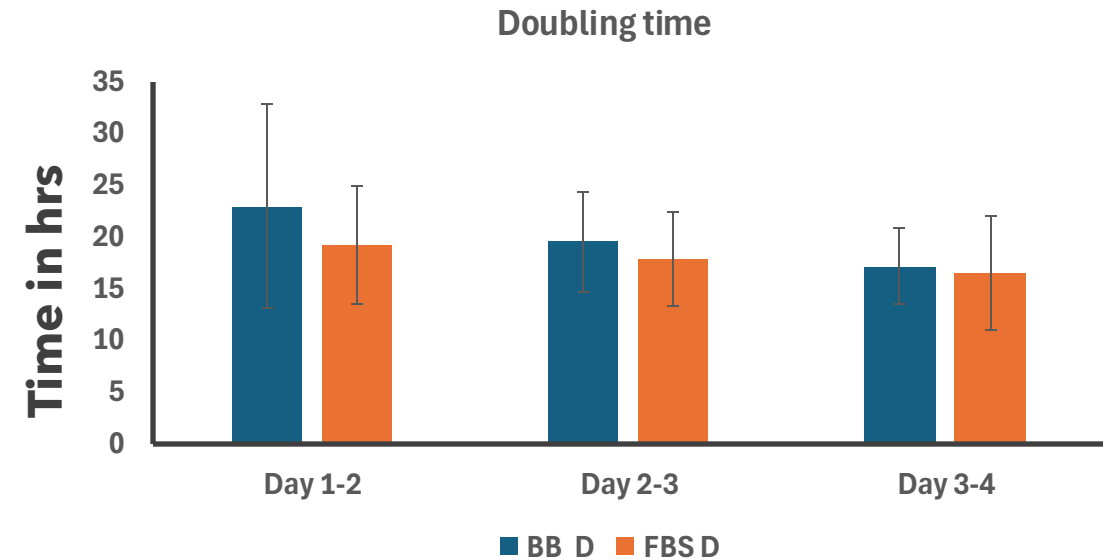
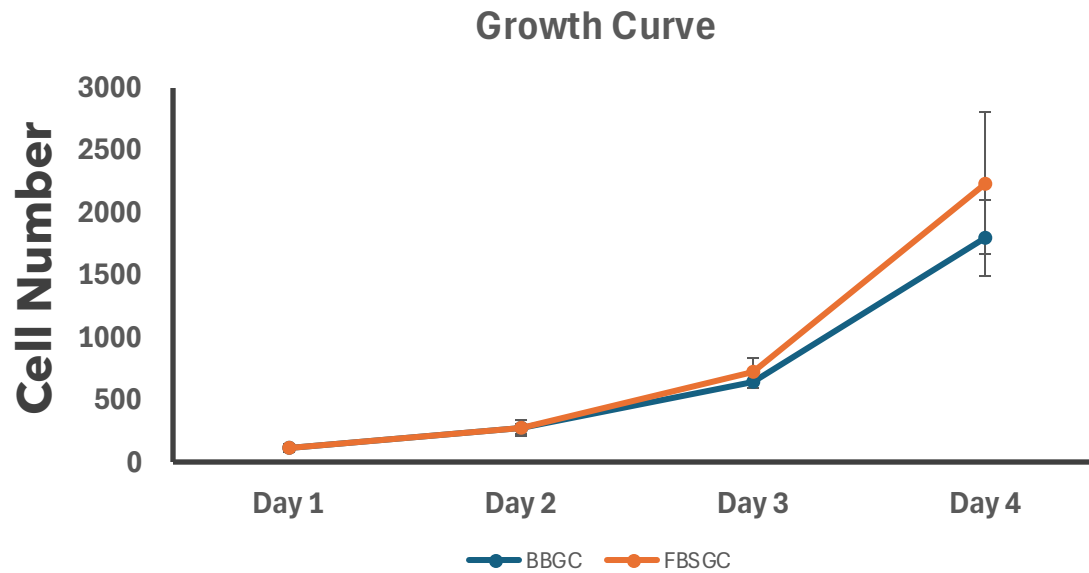
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SFM-Serum Free Medium, FBS-Fetal Bovine serum

5) Booster is comparable with FBS across ten generations with consistent functional capability



5) Growth kinetics studies show Booster & FBS are similar across 4-day period with similar doubling time



6) IPs Needed for Making Booster

- The cell-based serum replacement is a ground breaking alternative to fetal Bovine serum, for which we have IP and exclusivity (WO2024005714A1 A8 WIPO (PCT) ("Directionally-specific magnetic modulation of the cell secretome for medical and commercial applications")
<https://patents.google.com/patent/WO2016178631A1/pt>
- Collapsible cell-encapsulating scaffolds to enhance magnetically-induced secretome stabilisation and potency for which we have IP and exclusivity (<https://patents.google.com/patent/WO2025212041A1/en>)

7) Scalability and Manufacturing

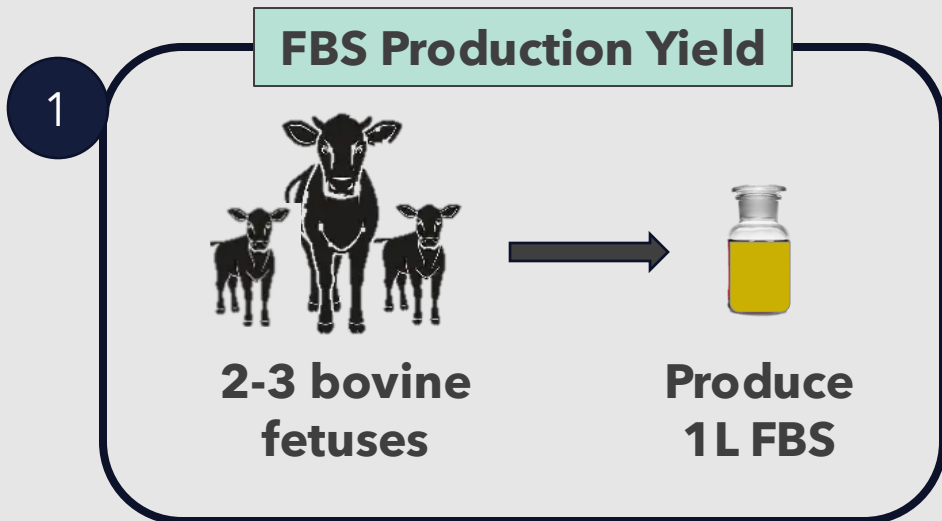
Six different factors considered

- 1 FBS Production Yield
- 2 FBS Production Time
- 3 Booster Production Yield
- 4 Booster Production Time
- 5 Cost Drivers
- 6 Scale Assumptions

7) Scalability and Manufacturing

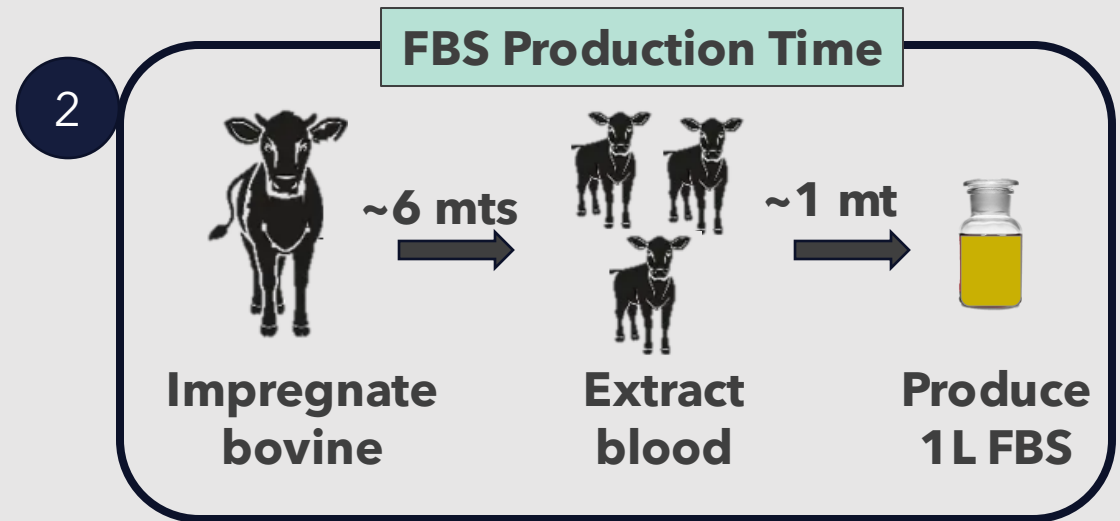
FBS Production Yield

- ~2 to 3 bovine fetuses are typically required to produce 1 litre of FBS
- As the demand for FBS increases, the number of fetuses increases.
- This poses scalability and supply chain challenges



FBS Production Time

- ~6 months from fertilization to blood extraction
- ~1 month for processing and QC testing



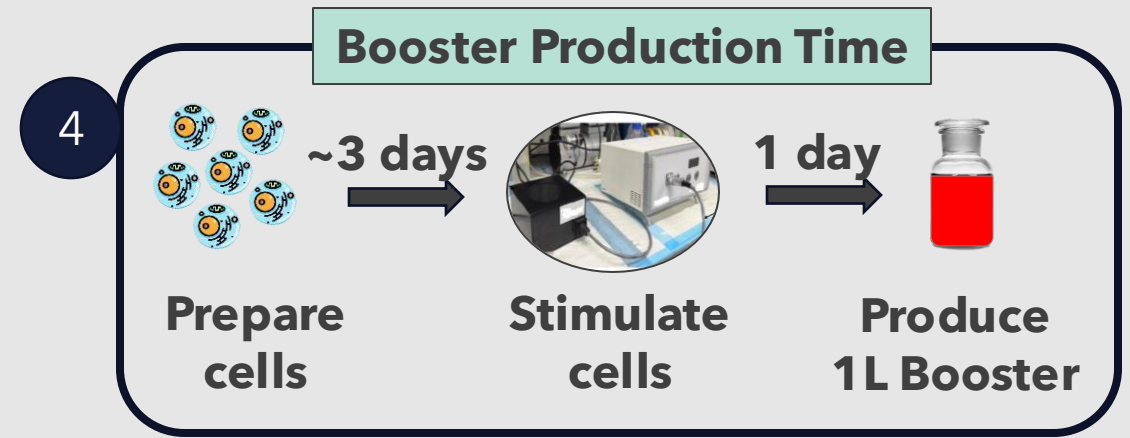
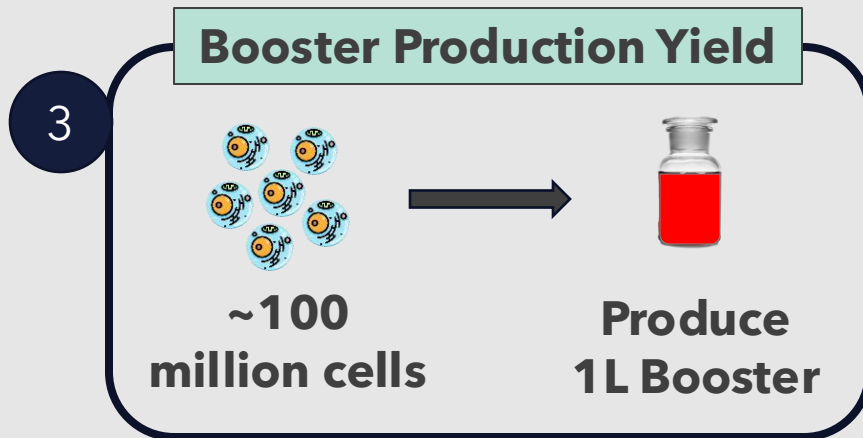
7) Scalability and Manufacturing

Booster Production Yield

- ~100 million cells are typically required to produce 1 litre of Booster
- As the demand for Booster increases, the number of cells required increases
- However, this poses minimal scalability and supply chain challenges because increasing cell numbers is a standard process

Booster Production Time

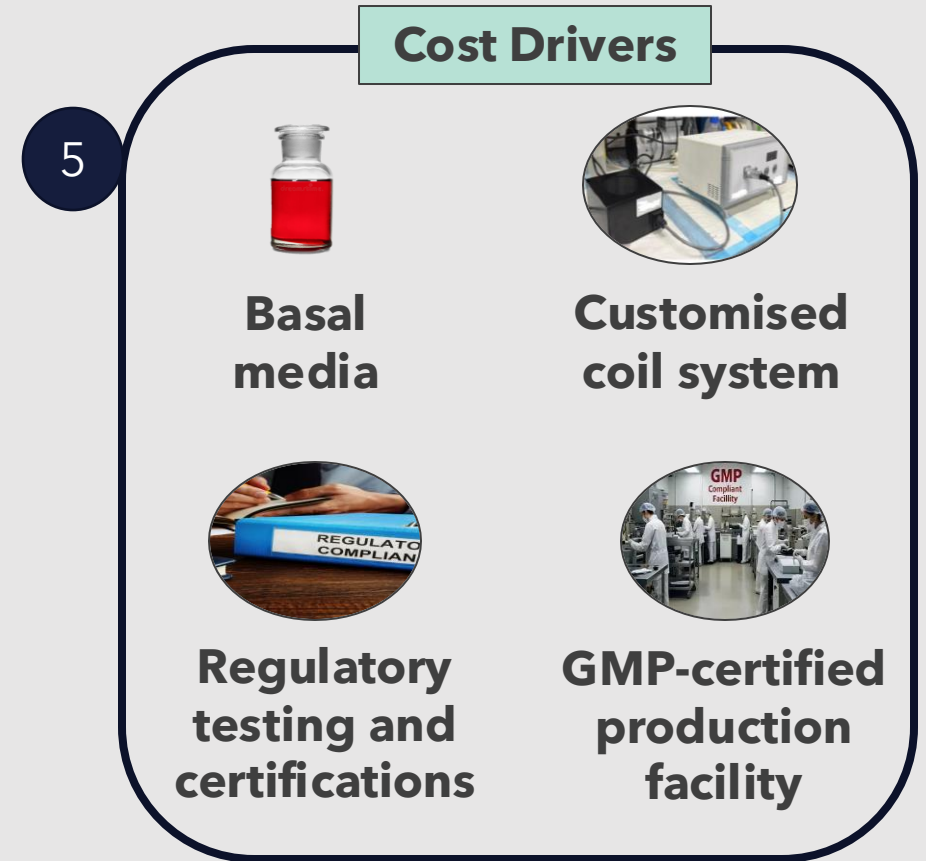
- Prepare cells
- 10 mins of stimulation
- <1 day for processing and QC testing



7) Scalability and Manufacturing

Cost Drivers (High Level)

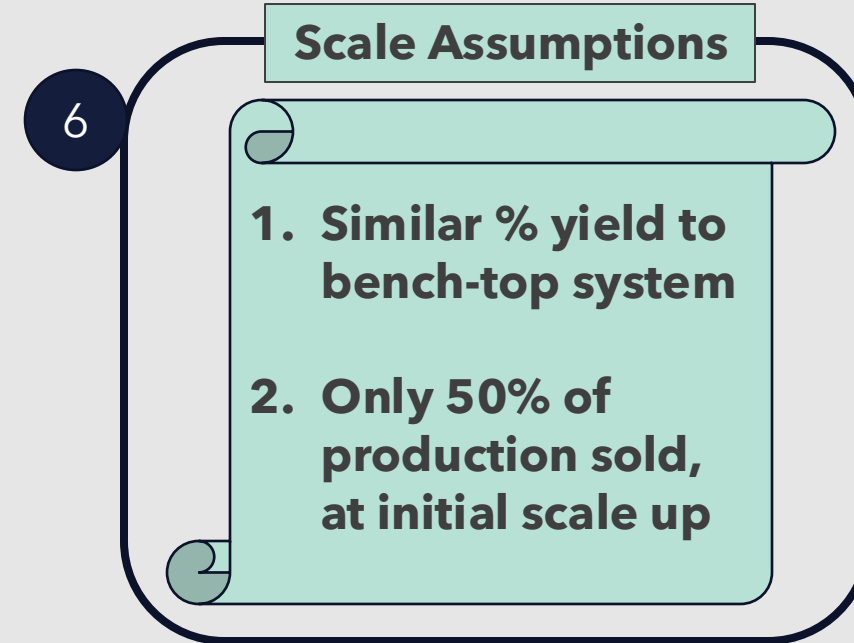
- Basal media
 - Large quantities required for preparing cells
 - In conversation with large-scale manufacturers for reducing costs
- Customised coil systems
 - No off-the-shelf options available
- Regulatory testing and certifications
- GMP-certified production facility
 - For reaching larger companies, GMP-compliance required
 - Achieved by developing in-house facility or outsourcing production



7) Scalability and Manufacturing

Scale Assumptions

- Similar percentage yield to bench-top system
 - Larger coil systems required as we scale-up
 - We assume output yield % will be similar to output of bench-top system
 - Efficiency of coils systems in stimulating cells expected to be consistent
- Only 50% of production sold
 - A conservative approximate for the initial scale-up process



8) Quality & Regulatory Pathway

Animal-free and Xeno-free positioning

BOOSTER is currently being positioned as a research-use only (RUO) cell culture supplement designed to replace without relying on animal-derived input. We are working towards GMP Compliance Grade.

Why FBS creates friction

- Animal-derived input raises ethical and sourcing concerns
- Batch-to-batch variability can create inconsistent outcomes
- High cost and limited supply make scale harder
- Supply chain dependencies create operational risks
- Risk of cross-contamination affects product reliability

Why BOOSTER is better positioned

- Animal-free and xeno-free composition.
- A more controlled biological input supports greater consistency
- Better aligned with scalable manufacturing and customer expectations

No animal input

More controlled

Built to scale

Animal-free by design

Positioning shift

A simpler path from variable serum to a more controlled biological input

This supports a simpler starting point for quality and regulatory buildout.

8) Quality & Regulatory Pathway

Regulatory alignment

- 1. Intended use** Research Use Only keeps the use case clear, and claims disciplined
- 2. Quality evidence** CoA, batch records, mycoplasma, pH, performance testing and shelf-life work
- 3. Pilot validation** Structured BOOSTER vs FBS pilots generate real-world performance evidence
- 4. Manufacturing backbone** GMP-grade capability strengthens scale without changing today's RUO frame(for next stage)



What this means in practice

- Claims stay anchored to evidence
- Every lot is documented and traceable
- Quality depth expands in line with scale, not ahead of it
- Clear path from early credibility to stronger operational readiness

Disciplined today. Scalable tomorrow.

8) Quality & Regulatory Pathway

GMP readiness roadmap

Today - foundation

- RUO positioning and claims discipline
- Batch records, lot traceability and baseline QC
- Structured pilot support already in place



Next 6 months - system buildout

- Approval workflows and training matrix
- Defined release checks and QC thresholds
- Early supplier screening and stronger data handling



Scale-ready - GMP-grade capability

- Transition toward GMP-grade manufacturing capability
- CAPA, deviation and release systems added as scale grows
- Certification readiness when justified by customer need

Quality systems are built progressively in line with how the business scales.

8) Quality & Regulatory Pathway

Research Use Only (RUO) – No formal certifications required at this stage

Defined quality control framework in place

- CoA generated for every batch
- Batch records and controlled documentation
- Mycoplasma, Endotoxin and Sterility Testing
- Basic release parameters: pH, appearance

Structured for near-term scale

- QMS-lite implementation underway
- Supplier traceability and material documentation
- Stability studies and claim substantiation

Scale-stage pathway

GMP-aligned manufacturing, ISO 9001 when scale justifies it, and application-specific regulatory pathways



NOW RUO + batch-level QC
COA | Mycoplasma | Endotoxin | Sterility

NEXT QMS-lite + traceability
Stability + supplier controls

LATER GMP-aligned manufacturing
ISO 9001 when scale justifies it

8) Quality & Regulatory Pathway

Risks + mitigation

The key operating risks are visible early, and each one has a specific control plan behind it.



Approach

Identify risk early and put controls in place early

Risk

Mitigation already planned or underway

Product claims

Use RUO wording and formal claim review so communication stays anchored to data

Product consistency

Track every batch through records, traceability and BOOSTER vs FBS pilot comparison

QC maturity

Expand from baseline tests into clearer release thresholds and broader analytical depth over time

Scale-up execution

Build GMP-grade capability over the next 6 months rather than trying to leap there at once

Supplier dependency

Introduce supplier screening and controlled sourcing before volumes become meaningful

9) Future Potential

Platform Expansion Beyond FBS

- Skincare: EV-based regenerative actives
- Therapeutics: cell-derived signalling for targeted modulation
- Biomanufacturing: optimised growth systems across cell types

Building a Predictive Biology Engine

- Systematically generate data on:
 - Cellular responses to stimulation
 - EV composition under defined conditions
- Move from trial-and-error biology → predictable outcomes

9) Future Potential

AI-Driven Optimisation Layer

- Train models on:
 - EV profiles
 - Cell performance metrics
- Enable:
 - Rapid formulation design
 - Cell-type specific optimisation
- Aligns with the direction of AI enabled biology, which is the direction biology is heading

10) Competitive Technical Landscape: Existing Alternatives

Recombinant Proteins

- Defined growth factors and cytokines
- High control, but limited complexity
- Expensive and often cell-type specific

Hydrolysates

- Plant or yeast-derived nutrient mixtures
- Cost-effective, partially defined
- Inconsistent performance across applications

Human Platelet Lysate (HPL)

- Human-derived growth supplement
- Strong performance in specific cell types (e.g. MSCs)
- Donor variability, regulatory and scalability challenges

10) Competitive Technical Landscape

Parameter	FBS	Recombinant Proteins	Hydrolysates	HPL	Cellivate
Performance	High (broad)	Moderate-High (narrow)	Moderate	High (specific)	High (targeted + broad potential)
Consistency	Low	High	Moderate	Low-Moderate	High (designed output)
Scalability	Limited	Limited (cost-driven)	High	Limited	High (cell-based production)
Complexity (biological relevance)	High (undefined)	Low	Moderate	High	High (functional signalling)
Cost trajectory	Increasing, supply constrains	High	Low-Moderate	High	Decreasing with scale (expected)

- FBS is the benchmark—but deeply flawed | Alternatives could solve pieces of the problem, not the whole
- Cellivate to unify performance + scalability + control

10) Competitive Technical Landscape – Some References

FBS – Performance, Variability, Ethics

- van der Valk et al., *ALTEX*, 2018
- Brindley et al., *Biotechnology Advances*, 2012
- Gstraunthaler et al., *ALTEX*, 2013

Recombinant & Serum-Free Systems

- Gordon Sato & Hajime H. Barnes, *Analytical Biochemistry*, 1980
- Butler, *Applied Microbiology and Biotechnology*, 2005

Hydrolysates

- Choi et al., *Biotechnology and Bioprocess Engineering*, 2006
- Franek et al., *Biotechnology Progress*, 2000

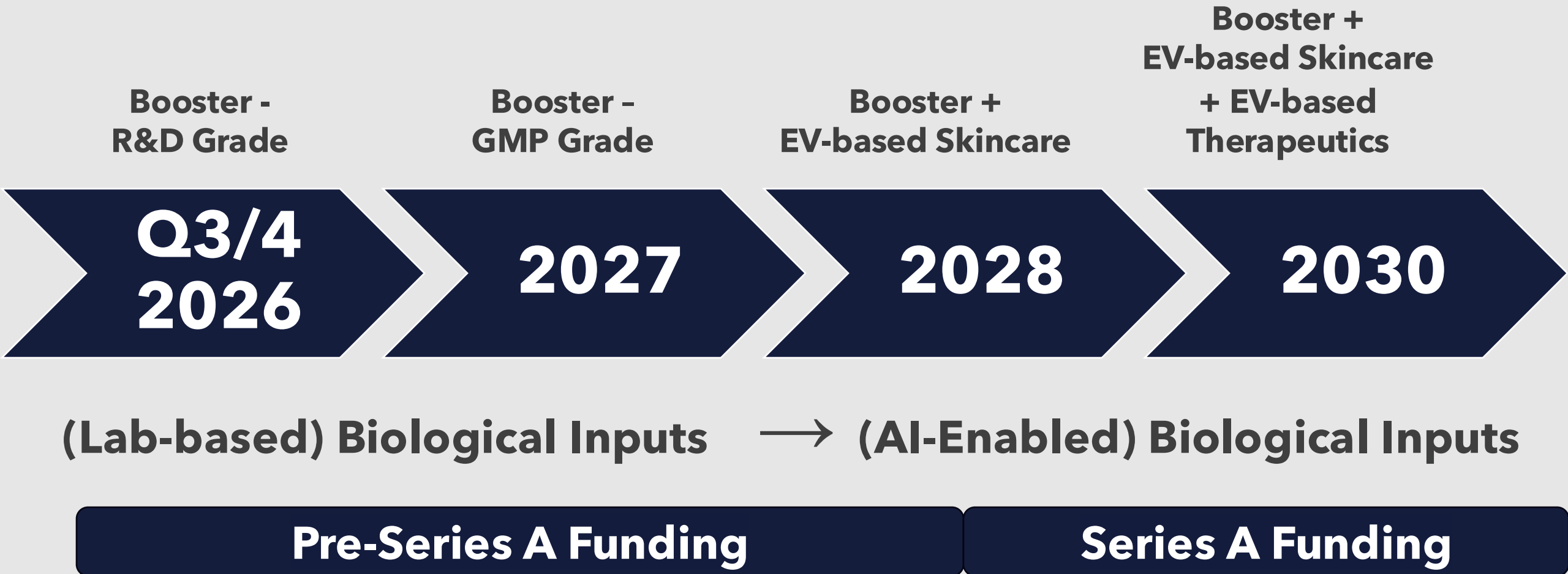
Human Platelet Lysate (HPL)

- Burnouf et al., *Transfusion Medicine Reviews*, 2016
- Hemeda et al., *Cytotherapy*, 2014

Serum-Free & System Comparisons

- Yao & Asayama, *International Journal of Molecular Sciences*, 2017

11) Technical Roadmap: Product Expansion Vision



11) Technical Roadmap

Key Experiments

- Head-to-head validation vs Fetal Bovine Serum across:
 - HEK, CHO, CarT Cells and stem cells (priority systems)
- Dose-response and optimisation studies
- Mechanistic studies:
 - EV characterisation (size, concentration, cargo)
 - Correlation: EV profile → cell performance
- Application-specific validation:
 - Viral titre (biopharma relevance)
 - Differentiation / proliferation assays

11) Technical Roadmap

Data Milestones

- Reproducible performance across multiple batches
- Defined input → output relationships:
 - Stimulation conditions → EV output → biological effect
- Build structured dataset for:
 - Cell-type specific optimisation
 - Skincare
 - Future AI/model development
- Internal benchmarking database vs industry standards

11) Technical Roadmap

Scale Milestones

- Optimised production protocol (yield + consistency)
- Pilot-scale manufacturing runs
- Initial cost-of-goods (COGS) model reduction
- Service orders already secured
- Early GMP-readiness planning (process + documentation)

11) Technical Roadmap

Validation with Partners

- Ongoing trials with:
 - Biopharma / vector production groups
 - Cell therapy developers
- Generate third-party validation data (critical for credibility)
- Convert pilots → paid evaluations → repeat users
- Strategic collaborations for:
 - GMP transition
 - Application-specific co-development

12) Why Cellivate Wins – Confidence from Technology to Market



**Is there
a need?**

- Product-Market-Fit established



**Can we
build it?**

- Developed prototype
- Successful validation in studies

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Scaling

- Yes
- This funding round

12) Why Cellivate Wins

Performance Edge

- Matches or exceeds Fetal Bovine Serum across key assays
- Maintains biological complexity through EV-mediated signalling
- Broad applicability across multiple cell types

Scalability Advantage

- Decoupled from livestock and donor supply chains
- Controlled, cell-based production system
- Favourable cost trajectory with scale

12) Why Cellivate Wins

Data Moat & Defensibility

- Proprietary dataset linking:
 - Stimulation conditions → EV profiles → cell outcomes
- Compounding data advantage over time
- Hard-to-replicate system requiring:
 - Biological know-how
 - Process optimisation
 - Integrated platform

Platform Extensibility

- Single platform, multiple verticals:
 - Biopharma
 - Cell therapy
 - Skincare



12) Why Cellivate Wins - Regulatory & Industry Alignment

1) Inevitable shift away from animal-derived inputs

Global regulators are increasingly moving toward defined, xeno-free systems to improve safety, traceability, and reproducibility in advanced biologics

2) Eliminates core compliance risks inherent to serum

Removes:

- Batch variability
- Contamination risk (adventitious agents)
- Traceability limitations
 - Key barriers in cell therapy, biologics manufacturing, and future GMP standards

3) Aligned with next-generation manufacturing paradigms

Enables transition from: Empirical, variable inputs → controlled, standardised biological systems



12) Why Cellivate Wins - Regulatory & Industry Alignment

4) Current alternatives are incremental, not transformational

- Recombinant proteins, hydrolysates, and platelet lysate are attempts to approximate serum:
 - Either oversimplify (defined but incomplete)
 - Or replicate variability (complex but inconsistent)

5) Cellivate's approach is fundamentally different

- Moves beyond "replacing serum" → redefining the input layer
- Uses cell-derived signalling (EVs) to deliver:
 - Biological relevance
 - Consistency
 - Scalability



Revolutionising Cell-Based Solutions for a Sustainable Future

Dr. Viknish Krishnan-Kutty

viknish@cellivatetech.com

+65-97356-007

